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1. INTRODUCTION

The **Diagnexia Validation/Verification Plan** for United States (US) is to evaluate the process (aka “Test method”) for Diagnexia Operational workflows that support patient clinical diagnostic testing, operational service workflows that support clinical diagnostic testing, &/or telepathology workflows. This aligns to College of American Pathologist (CAP) and Clinical Laboratory Improvement Amendment (CLIA) requirements.

2. PURPOSE

This plan will include test case steps to support the following:

- Intended Use in a CLIA lab environment or equivalent
- Meet CLIA / (CMS) and CAP requirements for clinical diagnostic intended use
- FDA Laboratory Developed Test, *when applicable*

Any testing performed under this plan will be summarized in the [DMS-108566](#), *Diagnexia Validation Summary Report - Clinical USA* to ensure the Laboratory Medical Director approves any testing for clinical diagnostic patient testing.

To establish performance characteristics for a new (validation) or updated (verification) clinical diagnostic test method, a protocol based on scientific methods and good laboratory practices is executed. Scenarios whereby a validation (or verification) may be needed are but not limited to:

- Upon each software feature release or MVP (minimal viable product) release - [DMS-8333 \(SSDLC\)](#) or [DMS-77274 \(ADLC\)](#)
- Individual equipment Installation Qualification (IQ), Operational Qualification (OQ), or Performance Qualifications (PQ) - [DMS-8475](#)
- New or changed operational workflows (*e.g. Standard Operating procedure updates or Engineering change orders*) - refer to [Appendix A](#)

3. SCOPE

This protocol applies to the workflow validation supporting Diagnexia’s clinical laboratory, **Naperville (USA)** or remote configurations that are linked to the laboratory operations (*e.g. telepathology or customer digital integrations*). This may also include any referral or customer laboratory operating under the CLIA certificate at the Naperville, IL (USA) location.

It may include the following, but not limited to:

- Comparative test method intended for client submitted patient samples (*scanned glass, digital data, or peripheral configurations*) - [Appendix A](#)

- Technical &/or laboratory test methods based on Diagnexia’s internal quality control procedure, ([DMS-8094](#))
- Whole Slide Imaging (WSI) scanner qualifications or GLP fit for purpose / commissioning testing
- Software deployments (*Diagnexia off the shelf configurable software*) - [DMS-8333](#)
- Workstation components (*e.g. monitors, peripheral devices*)

3.1 **Risk Assessment**

A project level risk assessment is to be performed to evaluate which parameters require a test method process performance evaluation; refer to [DMS-105073](#).The Risk evaluation will justify testing on higher risk changes or new features, while lower risk items may be accompanied by peer review literature or acceptance of risk criteria by the laboratory medical director or equivalent role (e.g. Chief Medical Officer) in the project. A final medical attestation must be obtained from the Laboratory Medical Director when regulatory compliance requires it.

3.2 **Regulatory**

To meet Regulatory requirements for Whole Slide image system(s) & clinical diagnostic intended use within diagnostic workflows, the following requirements must be met:

Regulatory Agency	Requirement Description
FDA	Any equipment used for clinical diagnostic use must be commissioned using IQ, OQ, & PQ methods. Intended use of the Whole slide image system or any Laboratory Developed Tests must accompany an intended Use statement with supportive testing &/or Peer reviewed literature supporting validations.
CLIA / CMS	Test Methods must support: Analytic sensitivity, specificity, precision, and accuracy - 42 CFR 493.1253 or accompany equivalent justification.
CAP	The laboratory medical director must approve clinical test methods & qualified equipment before the intended use for clinical diagnostics may be used in patient testing. An attestation statement is recommended as: “ <i>I have reviewed the verification (or validation) data for the performance specifications listed below for the (insert instrument/test name), and the performance of the method is considered acceptable for patient testing</i> ”.

This Policy has been created as part of this commitment and is available to all directors, employees, officers, interns and contractors within Deciphex group of companies including at Deciphex Limited, Diagnexia UK Limited, Deciphex Inc d/b/a Diagnexia US and Diagnexia Canada Limited (“collectively the “ Deciphex Group” and individually a “Company” for

purposes of this Policy) and any other location where the company carries our services or operations. The Policy applies to all personal data that Deciphex Group holds and processes.

The policy applies to:

- Deciphex Group staff
- Company Board and committee members
- All contractors, suppliers and other people working on behalf of Deciphex Group

4. ROLES & RESPONSIBILITIES

ROLE	RESPONSIBILITY
Regulatory Affairs, (author)	Provide requirements and auditing of records.
Quality assurance	Provide Validation & GLP guidance. Ensure procedures, records, and support Standard Operating Procedures are aligned to requirements.
Sr. Director of Lab Operations	Provide resource support for testing, environment conditions to support testing, and applicable delegated authorization for testing.
Chief Medical Officer	Organizational support, testing, and review expertise for clinical validity and applicability in the testing approach, execution, and review.
Laboratory Medical Director	Clinical review and attestation in context of the Medical Laboratory Duties, outlined in DMS-8370 & CAP.
Independent reviewer	Person(s) outside the Validation team for unbiased review of information.

5. PROCEDURE

5.1 Pre-requisites - information may be required prior to testing, refer to [DMS-xxxxx](#).

Description	Examples
Software / Data	• Software version build ID • Software URL address, if it is a validation test site or production site • Physical location where the software is accessed • Data source: test data, production data - note: all data should be anonymized per DMS-8378 Diagnexia Work Instruction to Pseudonymize Data Prior to Scanning • Integration: Host site location for data
Internet speed	<i>if relevant to testing</i>
Equipment	• Location of the physical equipment • Unit Serial Number (SN#) • Manufacture's regulatory label, when relevant • Peripherals: Client Personal computer / device / workstation SN#, make, model
Associated control documents	• Software release notes • Automation Development Lifecycle Release notes • Associated equipment's installation, calibration, or preventative maintenance records • Reference Quality documents: • Engineering Change order(s), if relevant to a change control • Corrective Action Preventative Actions, Incident investigation, or Fresh Desk ticket # • Testing protocol, Standard operating procedures, Training records

5.2 System Description

5.2.1 Architecture - Refer to [DMS-72596](#) describes the architecture that is supported by this Validation Plan.

The Diagnexia telepathology software, hardware, and systems in United States (US) clinical operations generate diagnostic-quality digital images of glass slides containing formalin-fixed paraffin-embedded (FFPE) tissue samples. This includes slides stained with haematoxylin and eosin (H&E), immunohistochemistry (IHC), and other specialty stains used in clinical pathology review. The workflows evaluated will focus on pre-analytic, analytic, and post analytic workflow steps that support the WSI images captured by the digital scanners.

The workflows qualified under this protocol are a cumulative representation of ADLC ([DMS-77274](#)) & SSDLC ([DMS-8333](#)), and AI supportive and software code solutions designed in premise architecture and cloud based architecture. The installation (IQ) and operational (OQ) configurations ([DMS-20398](#)) are Diagnexia software-specific and designed to be hardware-agnostic across the WSI scanner fleet used in Diagnexia’s US labs.

5.2.2 Training & Competency - All personnel executing this protocol must be trained and qualified in accordance with Diagnexia policies applicable to US operations. This includes:

5.2.2.1 [DMS-8354](#) – *Policy and Procedure for Personnel Competency Assessment*

5.2.2.2 [DMS-41059](#) – *Competency-Based Training for the Quality Control of Digital Images*

The Diagnexia system may be operated by laboratory personnel such as lab technicians, histotechnologists, biomedical scientists, or project managers who are trained on digital pathology workflows and slide scanning equipment. Specific training may be necessary prior to executing any testing protocols supported by this Validation plan. The training requirements will be outlined in the testing protocol / report, refer to [DMS-xxxxx](#).

This system is intended for use in CLIA-compliant environments and must be operated in accordance with local and federal safety, security, environmental, ergonomic, and regulatory standards for high-complexity clinical laboratories.

5.3 Protocol deliverables

A protocol is written by a qualified person (*someone other than the tester*) to design the validation study towards learning objectives. A test protocol / test case is to be drafted & may require pre-approval prior to testing. Refer to [DMS-xxxxx](#).

Deliverables	Description
Learning objective(s)	The protocol author outlines the hypothesis statement or the outcome objectives that the validation study is targeting
Requirements	Scientific Method, Quality,Regulatory requirements, Good Documentation Practices, 21 CFR Part 11 e-signatures
Sample size	Statistically relevant or justified sample size. 95% confidence interval is widely accepted
Known controls	A known positive or known negative control must accompany any test variable
Specificity - “miss”	Type I (false positive) - the tester identifies a difference between the test parameter when really there is no difference
Sensitivity - “false alarm”	Type II (false negative) - the tester fails to recognize a difference between the test parameter when there is a true difference

Deliverables	Description
Precision	Reproducibility – measurements remain consistent when tested under varying conditions (e.g., different operators, instruments, or times) - "correctness"
Accuracy	Repeatability—the ability to get the same result under identical conditions—(e.g. A single operator, sample, or instrument can repeat the same result multiple times) - “consistency”
Acceptance Criteria	DMS-8094 or established definition of pass & fail criteria
Deviations	If the tester must follow steps that are different than in the protocol, they must record the deviation. If addressed with a re-test, the re-test steps and evidence is collected and dispositioned within the protocol. A revised report may be necessary to report evidence identified.
Re-Test	A repeated test step or test step(s) may be executed to achieve an objective result. The tester will record the steps used to re-test. Both the original test and re-test will remain in the test report record.
Medical Attestation	CLIA and CAP require that the US Laboratory Medical Director provide an attestation if the validation is supporting clinical patient care. CAP: COM.40475 “I have reviewed the verification (<i>or validation</i>) data for the performance specifications listed below for the (insert instrument/test name), and the performance of the method is considered acceptable for patient testing.”
Approvals	Testers, Quality, US Laboratory Operations, and Independent reviewers at minimum. Medical Lab Director is needed prior to patient care testing.
Roles in the testing	• Tester - executes the protocol test steps, addresses deviation test steps • Approvers - approves test records with e-signature • Author - designs, writes, and approves test records • Reviewer - reviews and provides independent review of test records • US Laboratory Medical Director - required approval prior to patient testing

6. CHANGE MANAGEMENT

6.1 All software changes ([DMS-8333](#) *Diagnexia Policy for Secure Software Development Life Cycle*), including ADLC ([DMS-77274](#)) or configuration modifications, must follow the defined change management process as outlined in [DMS-19364](#) (*Procedure for Change Management*). Prior to rollout, release notes must be documented and approved in accordance with [DMS-20398](#)). If the change is “MAJOR” or has customer, business, or technical dependencies an ECO [Engineering Change Order] may be necessary to carry out the change. Refer the Change Management policy [DMS-8318](#) for clarification on when an ECO may be necessary.

6.2 Any equipment-level software updates must be completed and recorded by authorized Diagnexia personnel or designated vendor technicians. Refer to [DMS-8765](#) *Diagnexia Procedure for Equipment*.

7. EXCEPTIONS

No Exceptions are noted for the plan.

8. REFERENCES

Document ID	Document Title
DMS-xxxxx	Deciphex Validation Policy
DMS-xxxxx	Diagnexia -USA Laboratory Validation Verification Template
DMS-8765	Diagnexia Procedure for Equipment
DMS-72576	Diagnexia Master Security Document
DMS-8305	Diagnexia Policy and Procedure for Quality, Governance, and Risk
DMS-8333	Diagnexia Policy for Secure Software Development Life Cycle (SSDLC)
DMS-77274	Policy for Automation Development Life Cycle for Automation Products

Document ID	Document Title
DMS-69916	Diagnexia Work Instruction for Integration Configuration including roles and responsibilities
DMS-8338	Deciphex Procedure for Document Control
DMS-8318	Deciphex Policy for Change Management
DMS-8094	Diagnexia Policy for Quality Control
DMS-46957	North America Quality Policy
DMS-46958	North America Quality Manual
DMS-8354	Policy and Procedure for Personnel Competency Assessment
DMS-41059	Competency based training for the Quality Control of Digital Images
DMS-19364	Procedure for change management
DMS-8333	Diagnexia Policy for Secure Software Development Life Cycle (SSDLC)
DMS-20398	Diagnexia Work Instruction for Operations oversight of Software rollout
DMS-105073	Quality Risk Assessment for Diagnexia project
DCX-REG-04	North America Risk Register
DMS-108040	Deciphex Master Validation Plan for GLP Services
DMS-108566	Diagnexia Validation Summary Report - Clinical USA
DMS-8370	Diagnexia Procedure for Laboratory Personnel Requirement

Regulatory / Quality Assurance References	Description
College of American Pathologist (CAP)	GEN.42195, GEN.43022, GEN.50630, GEN.52920, COM.40475, (COM.40640)
Clinical Laboratory Amendment Act (CLIA)	42 CFR 493.1253 - Establish Performance Characteristics
CLIA Testing Guidance	https://www.cms.gov/regulations-and-guidance/legislation/clia/downloads/6064bk.pdf
Good Laboratory Practice	21 CFR Part 58
Food & Drug Administration (FDA)	21 Part 11 - electronic signature
FDA Cure's Act (law)	21st Century Cures Act - determines how to classify medical device(s)
FDA GUIDANCE: Medical Device Data Systems - (2022)	Discusses how the FDA does not plan on enforcing devices that only transfer data. The FDA considers “acquiring” data still a device (eg scanner) Refer to 21 CFR 864.3700
FDA GUIDANCE: General Principles of Software Validation - (2002)	Discusses how to validate Configurable Off-the-shelf-software (COTS) and includes change control risk based guidance
FDA GUIDANCE: Artificial Intelligence in Software as a Medical Device	Outlines the levels of AI. Because AI tools used in Diagnexia are Class A (safety), the tool is not making decisions; it will still rely on the human. The AI tools used in Diagnexia are sorting and classifying data to make the decision simpler for the human.
FDA Statistical Technicals & Sample plans (QIST)	21 CFR 820.250 & https://www.scribbr.com/statistics/normal-distribution/

9. DEFINITIONS & GLOSSARY

9.1. DEFINITIONS

TERM	DESCRIPTION
Commissioning	Commissioning is the methodical process of ensuring that all components of a project or system are installed, tested, and functioning according to specified requirements before it is put into service.
Fit-for-Purpose	A product, service, or system is capable of fulfilling its intended function or purpose effectively and reliably.
Intended Use / Purpose	Intended use is the specific purpose for which a product, device, or system is designed and marketed, as defined by its manufacturer in regulatory labeling and instructions for use (IFU).
Validation	Validation ensures that it fulfills its intended use and satisfies user needs.
Verification	Verification is the process of checking whether a product or system meets specified requirements.
Risk	Risk is likelihood & consequence impact, which can be either a threat with negative consequences or an opportunity with positive outcomes.
Qualification	Documented process of demonstrating that a process or equipment consistently operates within specified limits and produces results that meet predefined quality and performance standards.
Laboratory Developed Test - (LDT)	An in vitro diagnostic test that is designed, manufactured, and used within a single clinical laboratory certified under CLIA, and it is subject to the regulatory requirements for high complexity testing as defined by CLIA.
RUO - <i>Research Use Only</i>	Products or tests that are intended strictly for scientific research purposes and <u>not</u> for diagnostic or clinical decision-making, meaning they are not validated for use in patient care and lack the regulatory oversight required for medical testing.
IVD - <i>In Vitro Diagnostic</i>	Product, equipment, or test intended for the diagnosis of disease or other conditions from specimens taken from the human body, and its labeling must include the device name, intended use, manufacturer details, warnings, traceability (such as version or lot ids), and the statement “For In Vitro Diagnostic Use”.
510K - “cleared”	A 510(k) label must specify the device’s intended use, indications, warnings, manufacturer details, and demonstrate through supporting documentation (<i>including validation data</i>) that the device is substantially equivalent to a legally marketed predicate device and safe and effective for its labeled use. [Requires pre-market approval (PMA) filing with FDA]. <i>Sometimes called “cleared - device”.</i>

TERM	DESCRIPTION
IUO - <i>Investigational Use Only</i>	Product label means the item is intended strictly for use in clinical investigation or product testing, not for routine clinical diagnosis or patient management. <i>[Useful to have on field exposed software screens or products under evaluation]</i>
Proficiency Testing	Periodically tested samples and compared results with other labs to ensure the accuracy and reliability of their testing processes; successful participation is essential for demonstrating ongoing competency in clinical testing.
Sensitivity - Type II	Sensitivity is the probability that a test will correctly identify true positives (samples with the condition); thus minimizing Type II error (false negatives).
Specificity - Type I	Specificity is the probability that a test will correctly identify true negatives (samples without the condition); thus minimizing Type I error (false positives).
Accuracy - within	Refers to how close the test results are to the actual condition. Measured by repeatability of a single tester, sample, or instrument. <i>[Inter-variability or within variable].</i>
Precision - between	The degree to which a repeated measurement is reproducible between variables; measured by multiple testers, samples, or instruments. <i>[Intra-variability or between variables].</i>
Statistical Methods	Statistical methods are quantitative techniques used to collect, analyze, and interpret data in order to identify patterns, evaluate variability, and draw valid conclusions about phenomena or processes. <i>Often expressed as a confidence interval (CI) of 95% to indicate 2.5% Type I & 2.5% Type II error acceptance in a process.</i> NOTE: Adequate sample size is necessary, $n=29$ - Central Limit Theorem.
Qualitative	Qualitative measurements describe characteristics or qualities in non-numeric terms.
Quantitative	Quantitative measurements express information numerically for statistical analysis and direct comparison.
Test Method Transfer	Test method transfer in validation is the documented process of demonstrating that a validated analytical or test method can be successfully performed by a receiving laboratory with equivalent accuracy, precision, and reliability as at the originating laboratory.
Human Factor Engineering	The process of evaluating and verifying that a product, system, or device can be used safely, effectively, and efficiently by intended users under real-world conditions.

TERM	DESCRIPTION
Usability	Objective testing that intended users can safely, effectively, and efficiently achieve intended goals with a product or system under realistic conditions.
Acceptance Criteria	Acceptance criteria are predefined, objective, and testable conditions that a product, process, or system must meet to demonstrate successful performance. Example: Defining Pass/Accept or Fail/Reject based on pathologist scored images or slides.
Design of Experiment	(DoE) A structured approach for planning and conducting controlled tests to determine the identification and optimization of critical variables, thus ensuring consistent, reliable performance within predefined specifications.
Deviation	Unplanned or unexpected departure from an approved process, procedure, instruction, specification, or established standard during controlled activities such as testing.
Bug	Error or omission in the software code causing unexpected behavior (coding error)
Defect	Deviation from requirements or expected results found in the software (workflow error)
Black Box testing	The tester validates the functionality of the system without any knowledge of its internal code or structure.
White Box testing	A software testing method where the tester has knowledge of the internal logic, code structure, and implementation of the system.
Alpha testing	Internal investigational testing whereby the evaluators are part of the manufacturing organization.
Beta testing	External investigational testing whereby the evaluators are not part of the manufacturing organization.

9.2. GLOSSARY of ACRONYMS

ABBREVIATION ACRONYM	DESCRIPTION
CLIA	Clinical Laboratory Improvement Amendment
RCPATH	Royal College of Pathologists
CAP	College of American Pathologist
FD&C	Food, Drug, & Cosmetic Act
FDA	Food and Drug Association
IQ	Installation Qualification
OQ	Operational Qualification
PQ	Performance Qualification
CQC	Care Quality Commission
UKAS	United Kingdom Accreditation Service
ISO	International Standard Organization
PPQ	Process Performance Qualification
GCP	Good Clinical Practices
GLP	Good Laboratory Practices
GDP	Good Documentation Practices
PACS	Picture Archiving Communication System
MDDS	Medical Device Data Systems
GAMP / ISPE	Good Automated Manufacturing Practice (<i>International Society for Pharmaceutical Engineering</i>)

ABBREVIATION ACRONYM	DESCRIPTION
CSV	Computer System Validation
URS	User Requirement Specifications
FRS	Functional Requirement Specifications
ECO	Engineering Change Order
COTS	Consumer Off-the-shelf (<i>sometimes called “configurable off-the-shelf”</i>)

10. DOCUMENT APPROVERS

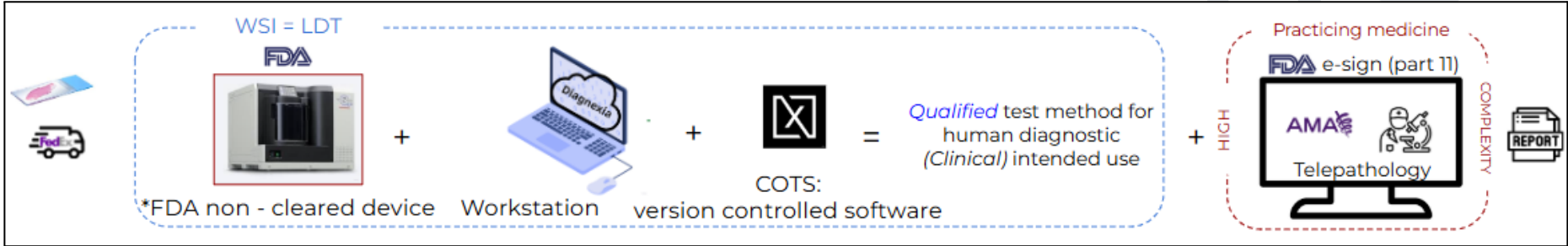
Approval role	Description
Laboratory Lead Technician	Responsible to uphold the Validation plan
Quality representative, US	Accountable to upload this Validation plan and associated quality principles outlined in the Quality Policy & Manual
Sr. Laboratory Director, N.A.	Accountable to support this Validation plan with operational resources (e.g. time, materials, testers)
Independent Reviewer	Person(s) outside the validation testing but has enough knowledge about Validation principles (e.g. GAMP, CSV, or laboratory methods)
Medical Lab Director CAP: DRA.10100	For CLIA & CAP compliance, the appointed US Medical Lab Director is needed prior to patient care testing.
Regulatory Affairs representative, US	Responsible to provide regulations or regulatory requirements supportive of US laboratory & validation compliance

11. REVIEW

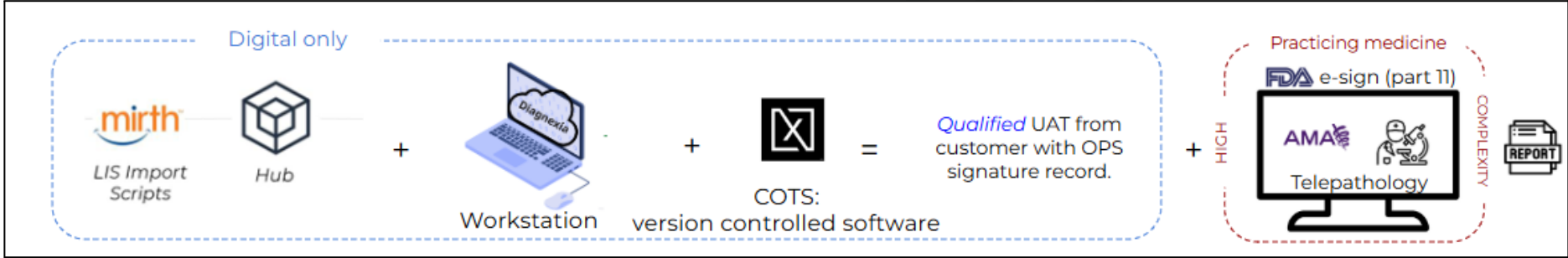
This document is required to be reviewed as per DMS-8338 Deciphex Procedure for Document Control to ensure that the procedure is relevant and does not require any change. An ad-hoc review may be required in response to any legislative, regulatory, contractual, organizational change or significant security incident or breach.

12. APPENDIX A: WORKFLOWS

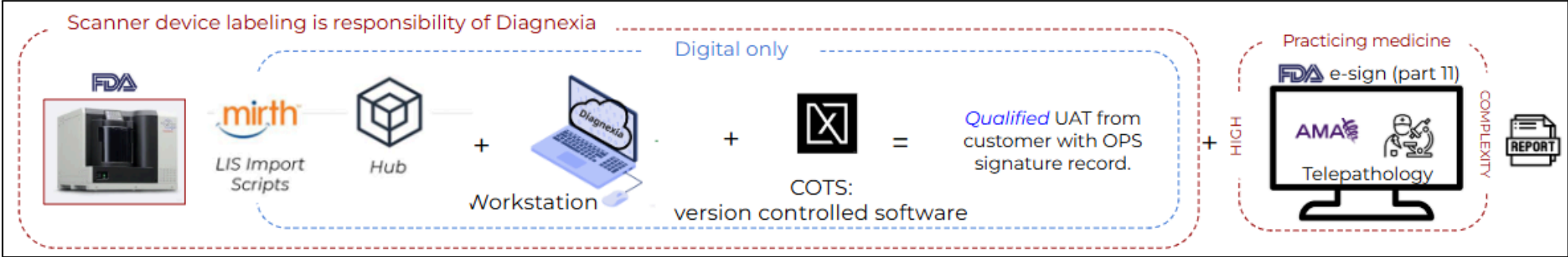
GLASS: Client ships glass to USA Laboratory; USA Laboratory processes patient samples then returns glass.



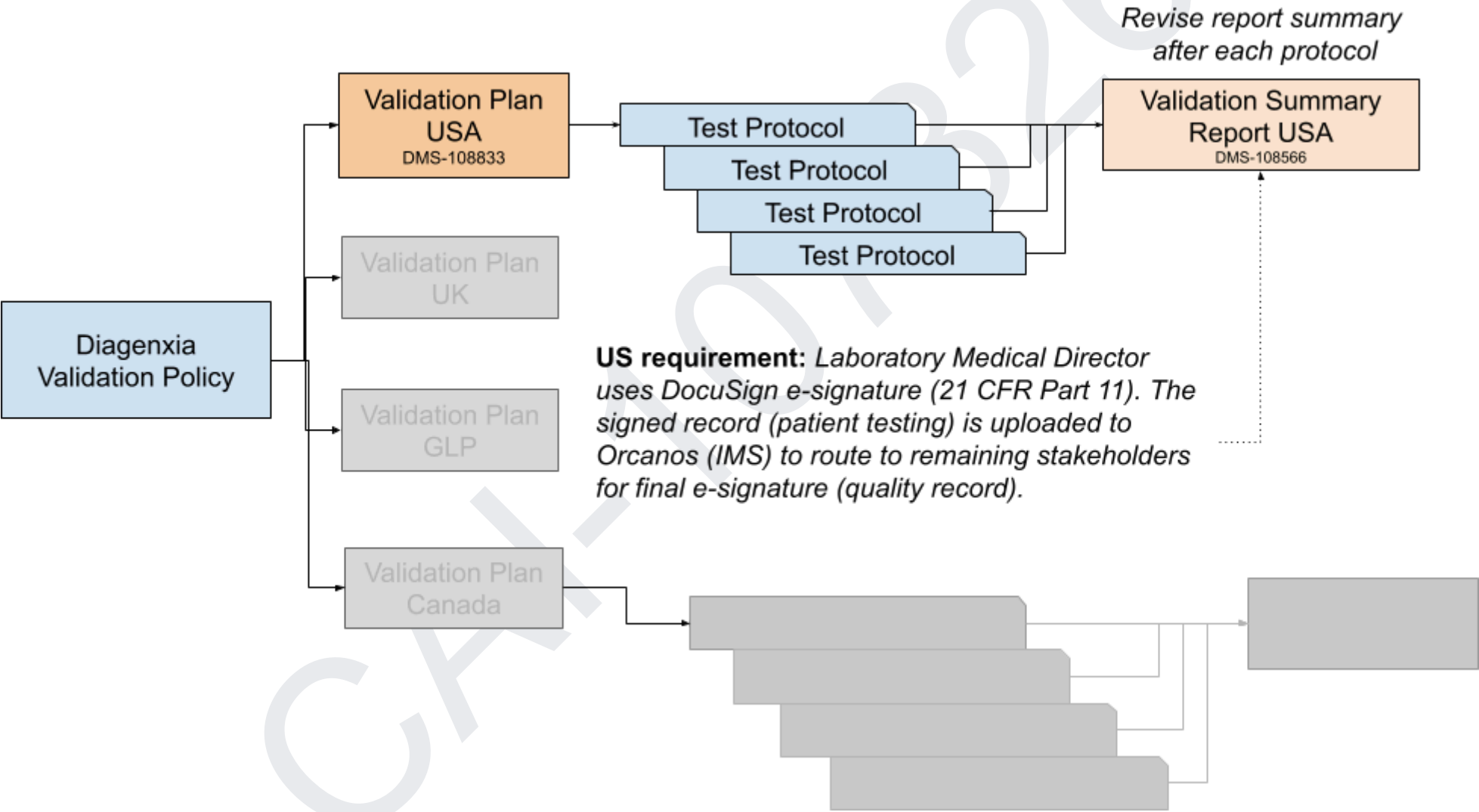
DIGITAL - 100%: Only data exchange occurring between client and USA operational laboratory



HYBRID: Combines scanner at client location with digital integrations - if Diagnexia “owns” the asset, Diagnexia is responsible.



13. VALIDATION RECORD APPROVAL LIFECYCLE



14. SIGNATURE MATRIX

21 cfr Part 11

[Name]	[Role]	E-signature	Comment / Note
	Author		With my signature as author of the document, I approve the content of the document.
	Tester #1		With my signature as an approver of the document, I approve the content of the document.
	Tester #2, Chief Medical Officer		With my signature as an approver of the document, I approve the content of the document.
	Independent reviewer		With my signature as an approver of the document, I approve the content of the document.